ContextWorld: Benchmarking In-Context Dynamics Adaptation in Latent Visual World Models

Abstract

Latent world models predict future states from visual observations and action sequences within a learned representation space. When environment dynamics vary, accurate prediction requires inferring hidden dynamic properties from interaction history. However, because models can exploit immediate visual shortcuts or regress to average dynamics to achieve low loss without history conditioning, conventional prediction accuracy fails to reveal genuine adaptation. We introduce ContextWorld, a benchmark designed to evaluate in-context dynamics adaptation in action-conditioned latent world models under strictly frozen weights. ContextWorld employs paired counterfactual evaluations—varying interaction histories while holding the current observation and future action sequence fixed—to concurrently quantify adaptation accuracy (whether predictions correctly distinguish between diverging future states) and history sensitivity (whether predictions respond to altered context and yield lower error than mismatched history). Spanning nine tasks across four environments, the benchmark provides separate training and evaluation splits with continuous interaction rollouts under differing dynamics. Our systematic evaluation of representative latent world models (LeWM, PLDM, and DINO-WM) yields three key findings: First, models trained under fixed dynamics make negligible use of interaction history across most tasks. Second, mixing multi-dynamics data improves history responsiveness, but the benefits of data scaling and the relative effectiveness of training strategies vary substantially across tasks. Third, under certain training schemes, models fail to effectively leverage history even on the training distribution, despite steadily decreasing training loss. These results reveal a notable mismatch between prediction-loss optimization and in-context dynamics adaptation, suggesting that standard prediction objectives do not reliably drive models to use interaction history and highlighting the need for objectives and benchmarks explicitly targeted at in-context dynamics adaptation.

1. Introduction

Visual world models learn environment dynamics to predict future states [19], with latent world models performing these predictions within learned representations of visual observations [20, 21]. A model trained under fixed dynamics can achieve high prediction accuracy without utilizing interaction history to adapt its predictions to current conditions. However, such predictions can fail when environment dynamics vary [33]. For example, visually identical objects can move different distances under the same push if surface friction differs. Interaction history provides evidence about these hidden physical properties. We investigate whether latent world models can leverage this contextual evidence to predict future states under varied dynamics *without updating their weights*—an ability we term *in-context dynamics adaptation*.

Physical prediction benchmarks evaluate whether observed interactions help models infer hidden properties or predict downstream outcomes governed by them [4, 32]. While benchmarks like Physion++ assess physical understanding via auxiliary readout classifiers, we introduce ContextWorld (Figure 1) to evaluate in-context dynamics adaptation directly within action-conditioned latent world models, leveraging their existing latent predictors without training separate classifiers for future events. The benchmark spans nine tasks across four environments, covering physical parameter variations and interaction rules, and provides separate training and evaluation data.

In its paired tasks, two dynamics conditions share the identical current observation and future action sequence, but produce diverging ground-truth futures. Interaction history is therefore strictly necessary to distinguish the two outcomes. Because a deterministic model that ignores history receives identical inputs and yields identical predictions for both conditions, it scores at most 50% on paired future selection when ties count as failures. This paired counterfactual design restricts history-agnostic heuristics, enabling us to concurrently evaluate two aspects directly in latent space: (1) adaptation accuracy, which measures whether a model's prediction correctly aligns with its corresponding true future rather than the alternative condition's future; and (2) history sensitivity, which measures whether predictions re-

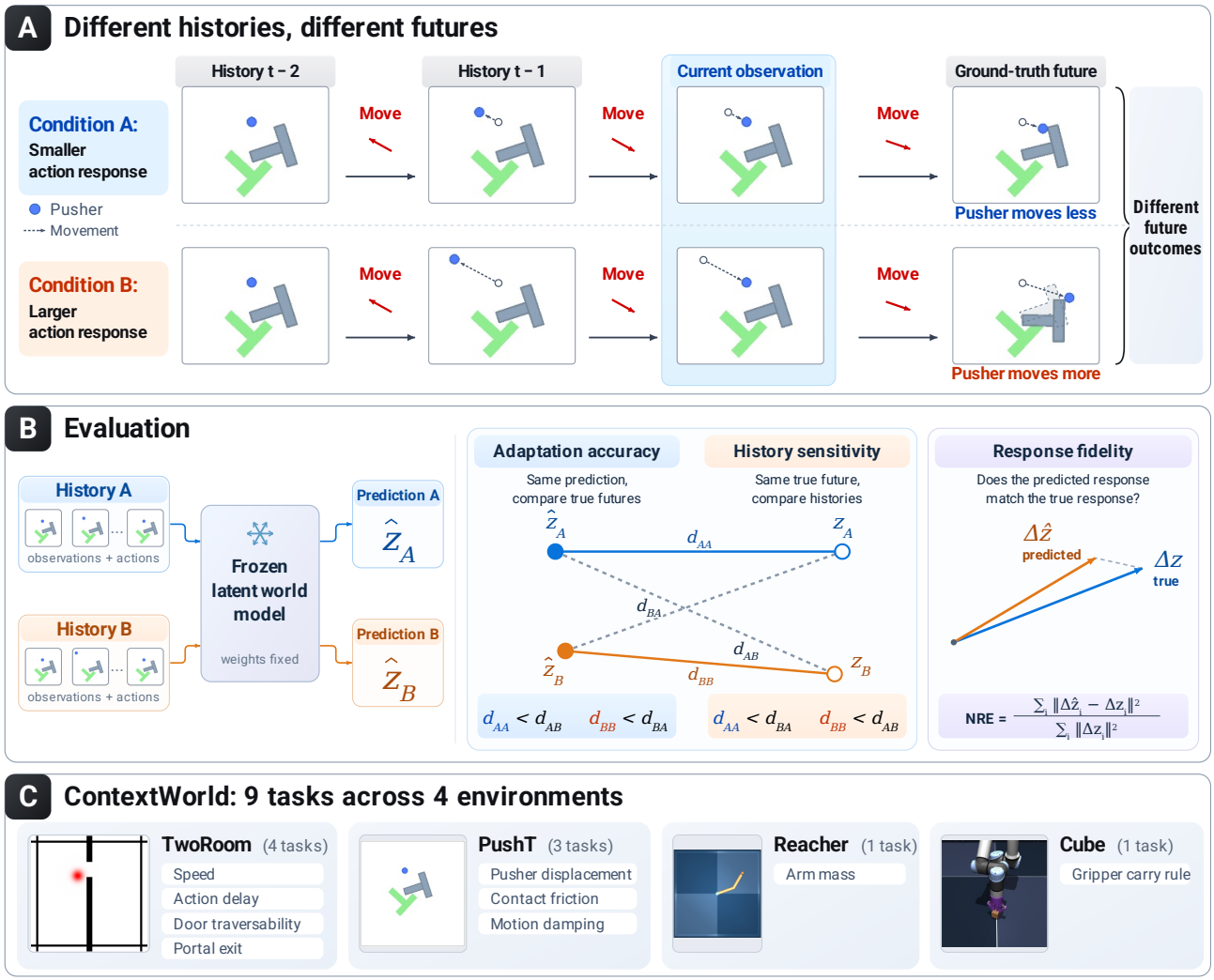


Figure 1. **ContextWorld overview.** (A) A recorded pusher-displacement pair with shared current observations and future actions. Red arrows show action directions; dashed arrows mark observed endpoint displacement. The translucent outline in the lower future panel marks the T block's current pose. These annotations are not model inputs; all scene images use the same crop of the original observations. (B) The frozen model produces latent predictions from each history. Three complementary checks evaluate paired tasks. Filled circles denote predictions and open circles denote encoded true futures. Accuracy and history sensitivity share one four-point diagram: accuracy compares true futures for a fixed prediction, whereas history sensitivity compares histories for a fixed true future. Here d_{ij} is the error between the prediction from history i and true future j; the inequalities show the desired ordering for both conditions. Response fidelity compares $\Delta\hat{z} = \hat{z}_B - \hat{z}_A$ with $\Delta z = z_B - z_A$, using the same schematic points. All latent-space geometry is illustrative, not a projection of measured model outputs. (C) Nine tasks across four environments.

liably respond to altered context, verified by substituting the history with that from an alternate dynamics setting to examine if matching history yields strictly lower error.

Using ContextWorld, we benchmark three representative latent world models: LeWM [24], PLDM [30], and DINO-WM [37]. We find that after training under fixed dynamics, these models make negligible use of interaction history across most tasks. To examine how this adaptation ability can be acquired, we mix ContextWorld training data with original rollouts and compare training from scratch against fine-tuning models pretrained under fixed dynamics. We further scale up the training data and examine the impact of freezing the visual encoder. Our experiments reveal that while data scaling improves history utilization on several tasks, the optimal model and training strategy vary substantially by task. For example, starting from the same pretrained weights, freezing PLDM's visual encoder improves its ability to distinguish future states under varying contact

friction, but reduces this ability under motion damping.

To understand why models often fail to leverage history, we analyze model predictions on the training distribution itself. Under certain training schemes, training loss steadily decreases or approaches convergence even while models fail to use interaction history to distinguish physical conditions. Because this failure occurs on the training data, it cannot be explained by out-of-distribution generalization alone, suggesting that standard prediction objectives may not inherently provide sufficient incentive for models to condition their predictions on interaction history.

Our main contributions are:

- **The ContextWorld benchmark and paired evaluation protocol.** We introduce a standardized framework featuring separate training and evaluation data across nine hidden-dynamics tasks. This paired design quantifies both adaptation accuracy and history sensitivity directly from action-conditioned latent rollouts.
- **Empirical findings on adaptation limits.** We benchmark representative models across data scales, training strategies, and visual encoder freezing schemes, demonstrating that optimal strategies are highly task-dependent. Furthermore, we show that training loss can steadily decrease even when models fail to leverage history on the training data, revealing a disconnect between standard prediction losses and in-context dynamics adaptation.

2. Related Work

Visual world models. Visual world models learn environment dynamics to simulate and predict future states. Early frameworks [19] combined learned visual representations with recurrent dynamics to support internal policy training. Subsequent methods advanced this paradigm through online planning (PlaNet [20]) and policy learning via imagined latent trajectories (Dreamer [21]). While these early approaches rely on pixel-level image reconstruction to learn representations, Joint-Embedding Predictive Architectures (JEPAs) [23] shift the training objective toward predicting target representations directly without reconstructing pixels. This principle has been applied to masked spatial regions (I-JEPA [2]) and extended to temporal video representation learning (V-JEPA [5]). To support robotic planning, V-JEPA 2 [3] introduces an action-conditioned predictor trained on robot interactions atop pretrained video representations.

Action-conditioned latent world models also differ in how they obtain their visual representations. DINO-WM [37] trains a predictive dynamics model on frozen pretrained features for planning from offline trajectories. In contrast, PLDM [30] jointly optimizes the visual encoder and the predictor using reward-free offline data. To stabilize joint training, LeWM [24] simplifies the objective to a latent prediction loss and a single representation regularizer. We benchmark these three paradigms to evaluate how structural choices influence in-context dynamics adaptation.

In-context adaptation. When environment dynamics differ between training and deployment, interaction history provides a signal for adaptation without requiring weight updates. Meta-reinforcement learning [6] leveraged this principle by training agents to adapt from limited experience across tasks. RL^2 [14] implicitly stores interaction history within a recurrent network's hidden state, while PEARL [29] infers a task representation from experience to condition the policy. Paralleling large language models, which learn from in-context examples without parameter updates [8, 13], this adaptation paradigm has been extended to visual world models.

Within world models, context guides predictions of action outcomes in new environments. OSWM [16] pretrains a transformer on synthetic dynamics, using observed state transitions as context to forecast futures. L2World [33] distinguishes using context to identify previously seen environments from learning new dynamics. It studies the effects of context length and training environment diversity through multi-step prediction errors and context perturbations. For robotic control, ICWM [34] employs task-agnostic interaction history to adapt policies to unseen camera viewpoints without weight updates, using ablations over correct, absent, and mismatched contexts to validate history reliance.

Benchmarks. To evaluate adaptation across environments, existing benchmarks introduce controlled variations in dynamics, appearance, or task rules. Meta-World [35] and Procgen [11] test generalization across manipulation tasks and procedurally generated levels. CausalWorld [1] and CARL [7] vary environment properties, while the XLand series (XLand-MiniGrid [26], XLand-100B [27]) evaluates policy adaptation from interaction history across distributions of tasks.

For physical prediction, several benchmarks focus on counterfactual reasoning and hidden property inference. CoPhy [4] asks models to predict outcomes after changes to a scene's initial configuration, using the original trajectory as evidence of hidden properties. Filtered-CoPhy [22] ensures the observed trajectory contains sufficient information for the counterfactual prediction. ComPhy [9] probes reasoning regarding hidden mass and charge through predictive and counterfactual video querying. Physion++ [32] tests whether interaction history helps predict future contact in paired trials that share the same current observation but differ in physical properties, assessing this understanding via auxiliary readout classifiers.

Beyond visual domains, a recent time-series benchmark [36] reveals that low prediction error can coexist with incorrect response directions when actions change. This

Table 1. Overview of the nine tasks evaluating in-context dynamics adaptation across four environments. History length h counts context observations (including the current observation) provided to the model. Sampling details and simulation settings appear in Appendices A and D.

Task	Environment	History length h	What history helps the model predict
Speed	TwoRoom	3	How far the agent moves under the same action
Pusher displacement	PushT	3	Whether the same action moves the pusher a shorter or longer distance
Arm mass	Reacher	3	How the same torque changes arm motion
Action delay	TwoRoom	7	When an action takes effect
Contact friction	PushT	3	How friction changes object motion during contact
Motion damping	PushT	3	How quickly object motion decays after contact ends
Gripper carry rule	Cube	3	Whether a closed gripper carries the cube
Door traversability	TwoRoom	3	Whether a visually identical door permits passage
Portal exit	TwoRoom	3	Where the agent emerges after entering the same portal

echoes our finding of a mismatch between prediction-loss minimization and true dynamics adaptation. A remaining question is when action-conditioned latent world models learn to leverage history for accurate prediction. ContextWorld addresses this gap through controlled comparisons across models, tasks, data scales, and training strategies.

3. ContextWorld Benchmark

ContextWorld is designed to evaluate in-context dynamics adaptation—specifically, whether models use interaction history to accurately predict future states conditioned on an action sequence. This section formalizes the prediction setting, describes how we construct the tasks and datasets, and explains the paired evaluation protocol and metrics.

3.1. Problem Formulation

To isolate in-context adaptation from weight updates, model parameters θ remain fixed during evaluation. The model receives exclusively visual observations and actions. For a target horizon τ, the model receives an interaction history H_t, comprising interleaved visual observations up to time t and their executed actions, alongside a future action sequence $A_{t:t+\tau-1}$. The prediction and its target are

$$\begin{aligned} \hat{z}_{t+\tau} &= f_\theta(H_t, A_{t:t+\tau-1}), \\ z_{t+\tau} &= e_\theta(o_{t+\tau}). \end{aligned} \tag{1}$$

Here, $o_{t+\tau}$ is the ground-truth observation at time $t+\tau$, and e_θ is the model's target encoder, which is also frozen. Predictions are evaluated directly against these encoded targets in latent space.

3.2. Task Design

We construct nine tasks across four environments by varying physical parameters and interaction rules. These environments are TwoRoom navigation [37], PushT [10], Reacher from the DeepMind Control Suite [31], and Cube from OGBench [28]. Figure 1 illustrates the task design, and Table 1 summarizes the tasks and their history lengths. Models are trained separately for each task, and DINO-WM keeps its pretrained visual encoder frozen.

Our main design is to test whether models can use interaction history to predict different outcomes when the current observation and future actions alone cannot distinguish the underlying dynamics. For six tasks—pusher displacement, arm mass, contact friction, motion damping, gripper carry rule, and portal exit—we construct pairs of samples that share the same current observation and future action sequence but have different interaction histories and ground-truth futures. Each sample is generated through a continuous simulation under fixed dynamics. We check relevant physical states, such as velocity and contact state, at the prediction point, since identical images do not necessarily imply identical physical states. For tasks where the two conditions cannot naturally share a starting position, we exchange their starting positions to prevent the initial images from revealing the dynamics condition. A model that ignores history therefore cannot distinguish the two futures within a pair (Appendix B).

The other three tasks examine history-based prediction through their respective task protocols. For Speed, we compare predictions conditioned on histories generated at different speeds, while keeping the current observation and query actions fixed. The model must achieve lower prediction error with the history corresponding to the target speed than with any alternative history. Evaluation includes seen speeds, interpolation, and extrapolation, while the main results use unseen interpolation speeds. For Action Delay, the model receives seven observation frames to infer action delays ranging from 0 to 10 simulator steps. Since the main prediction horizon is five steps, delays of 5–10 produce the same response within this horizon and are grouped into one

of six equally weighted response groups (Appendix C). For Door traversability, the model uses preceding interactions to predict whether a door is passable or blocked. Both rules appear during training, while Test uses 42 previously unseen door positions to evaluate generalization to new positions.

3.3. Dataset Construction and Splits

To evaluate models on unseen interaction rollouts, all nine tasks provide trajectory-disjoint Training, Development, and Test splits. Crucially, the Test set is reserved for final evaluation, while the Development set supports method and checkpoint selection. Across the six paired tasks, Action Delay, and Door Traversability, models are evaluated on unseen trajectories under dynamics conditions or rules represented in training. In contrast, the speed task assesses parameter generalization, with the main Test results reported for unseen speeds held out within the training range. Each paired task provides up to 32,768 training pairs, alongside 256 balanced counterfactual pairs in each evaluation split (Table 4, Appendix D).

3.4. Evaluation Protocol and Metrics

Prediction error alone does not establish in-context dynamics adaptation: models may reduce errors shared across dynamics conditions without predicting their differences. Conversely, selecting the correct future does not guarantee an accurate conditional response: Appendix F shows that a small predicted difference can still yield perfect future-selection accuracy. We therefore evaluate two aspects: Adaptation Accuracy and History Sensitivity. We additionally report Normalized Response Error (NRE) and Multi-Step Prediction Error (E/B) to assess conditional response and complete rollout errors.

Adaptation Accuracy (A_{future}). For each of the six paired tasks, let $i \in \{1, \dots, N\}$ index the N counterfactual pairs, with dynamics conditions A and B. Let z_i^A, z_i^B be the encoded ground-truth futures and $\hat{z}_i^A, \hat{z}_i^B$ the predictions from their respective histories. Write $d(x, y)$ for the mean squared error between latent representations x and y. A prediction is correct if it is strictly closer to the future under its own condition than to the alternative:

$$\begin{aligned} I_i^A &= \mathbf{1}[d(\hat{z}_i^A, z_i^A) < d(\hat{z}_i^A, z_i^B)], \\ I_i^B &= \mathbf{1}[d(\hat{z}_i^B, z_i^B) < d(\hat{z}_i^B, z_i^A)]. \end{aligned} \tag{2}$$

Here $\mathbf{1}[\cdot]$ equals one if its condition holds and zero otherwise. The percentage accuracy is

$$A_{\text{future}} = \frac{100}{2N} \sum_{i=1}^{N} (I_i^A + I_i^B). \tag{3}$$

Because the current observation and future actions are shared, a deterministic model ignoring history makes identical predictions for the two conditions and scores at most 50%; ties count as failures. Speed, action delay, and door traversability follow their own scoring rules (Appendix E).

History Sensitivity (History win rate). We test whether matching history reduces prediction error by keeping the current observation, future actions, and true future fixed while swapping the history. For paired tasks, define

$$\begin{aligned} J_i^A &= \mathbf{1}[d(\hat{z}_i^A, z_i^A) < d(\hat{z}_i^B, z_i^A)], \\ J_i^B &= \mathbf{1}[d(\hat{z}_i^B, z_i^B) < d(\hat{z}_i^A, z_i^B)]. \end{aligned} \tag{4}$$

The History win rate in percent is

$$A_{\text{history}} = \frac{100}{2N} \sum_{i=1}^{N} (J_i^A + J_i^B). \tag{5}$$

Unlike adaptation accuracy, which compares two targets for a fixed prediction, History win rate compares predictions from two histories against a fixed target; ties count as failures. The other tasks use their corresponding history comparisons (Appendix E).

Normalized Response Error (NRE). To assess whether predicted differences between conditions match the true differences, define $\Delta z_i = z_i^B - z_i^A$ and $\Delta \hat{z}_i = \hat{z}_i^B - \hat{z}_i^A$. We compute

$$\text{NRE} = \frac{\sum_i \|\Delta \hat{z}_i - \Delta z_i\|^2}{\sum_i \|\Delta z_i\|^2}, \tag{6}$$

where $\|\cdot\|$ is the Euclidean norm in latent space and the denominator must be positive. An NRE of 0 means the predicted difference is exact; a zero predicted difference gives NRE equal to 1. Differences cancel errors shared across conditions, so low NRE does not guarantee accurate

Table 2. Training schemes. Mixed combines original-environment and ContextWorld training windows at a 1 : 1 sampling ratio. Original → Mixed starts from an Original checkpoint, with Joint updating the visual encoder and Frozen Encoder keeping it fixed. DINO-WM uses a frozen pretrained visual encoder in all schemes.

Scheme	Start	Data	Encoder
Original	Initial	Original	Updated†
Mixed	Initial	Mixed	Updated†
Original → Mixed (Joint)	Original	Mixed	Updated
Original → Mixed (Frozen Encoder)	Original	Mixed	Frozen

Initial weights are random for LeWM and PLDM. DINO-WM uses a pretrained visual encoder and a random predictor. †DINO-WM keeps its encoder frozen in every scheme; its continued-training scheme is labeled Original → Mixed. Other modules remain trainable; native training losses are retained.

Table 3. **Adaptation metrics on the Test split.** Each cell lists Accuracy (%) ↑, History win rate (%) ↑, and NRE ↓ from top to bottom. Entries are means; run counts and standard deviations appear in Table 6. Speed uses the same scoring rule for Accuracy and History win rate. Bold marks the highest Accuracy and History win rate and lowest NRE per task, including ties at the displayed precision. NRE is measured in each checkpoint's target representation. A dash denotes no comparable checkpoint.

Model	Training scheme	Speed	Pusher displacement	Arm mass	Action delay	Contact friction	Motion damping	Gripper carry rule	Door traversability	Portal exit
LeWM	Original	58.04 58.04 0.896	48.96 28.32 1.018	50.07 51.63 0.977	—	49.93 50.13 1.001	50.00 48.63 1.007	52.21 73.76 0.909	50.33 50.94 0.996	50.07 51.95 0.990
	Mixed	98.19 98.19 0.042	50.78 79.10 0.974	85.74 89.84 0.045	97.70 99.18 0.013	50.00 49.41 1.000	50.00 48.24 1.004	50.00 49.80 1.000	99.83 **100.00** 0.041	50.78 76.37 0.922
	Original → Mixed (Joint)	96.89 96.89 0.055	91.80 99.02 0.215	87.30 93.36 0.038	—	50.39 54.30 0.999	50.00 49.61 1.001	50.00 52.34 0.997	99.83 **100.00** 0.038	50.39 66.80 0.934
	Original → Mixed (Frozen Encoder)	**98.44** **98.44** **0.030**	97.46 98.83 0.128	77.15 86.52 0.346	—	**86.13** **92.77** **0.429**	53.91 61.91 0.958	**76.37** **84.57** **0.271**	**100.00** **100.00** 0.036	77.93 94.34 0.264
PLDM	Original	48.41 48.41 0.922	45.57 9.38 1.058	51.82 66.99 0.921	—	50.00 49.09 1.014	50.26 50.52 0.947	50.78 57.42 0.937	50.06 51.50 0.997	50.07 51.50 0.996
	Mixed	96.78 96.78 0.084	56.25 68.16 0.835	**91.02** **93.55** **0.031**	**100.00** **100.00** **0.006**	50.00 48.63 1.001	58.20 55.86 2.435	49.80 50.78 0.997	**100.00** **100.00** **0.022**	72.27 79.10 **0.128**
	Original → Mixed (Joint)	98.33 98.33 0.047	59.18 73.44 0.740	84.77 90.82 0.073	—	50.00 46.48 1.020	**79.88** **81.84** **0.237**	50.20 54.30 0.982	**100.00** **100.00** 0.023	52.54 61.52 0.708
	Original → Mixed (Frozen Encoder)	96.44 96.44 0.062	**98.63** **99.80** **0.101**	81.84 89.26 0.213	—	73.63 83.59 0.595	54.49 64.26 0.867	52.93 57.81 0.606	**100.00** **100.00** 0.048	91.21 98.83 0.191
DINO-WM	Original	57.11 57.11 0.919	38.48 9.38 1.127	50.78 61.33 0.977	—	50.39 47.66 1.015	49.80 49.61 1.014	50.00 50.59 1.000	50.00 64.33 0.985	50.78 56.45 0.996
	Mixed	59.41 59.41 0.961	97.66 99.61 0.394	78.32 88.09 0.673	16.65 19.45 1.000	84.18 92.19 0.648	56.45 65.23 1.042	60.16 77.34 0.892	**100.00** **100.00** 0.217	**100.00** **100.00** 0.142
	Original → Mixed	84.78 84.78 0.411	—	—	—	—	55.27 61.13 1.111	62.89 80.27 0.871	—	—

complete predictions. Multi-condition tasks use their task-specific response comparisons.

Multi-Step Prediction Error (E/B). We evaluate open-loop autoregressive predictions at simulator steps 5, 10, 15, 20, and 25, without intermediate ground-truth observations. Let q index evaluation scenes; k, dynamics conditions; c, shared future action sequences; and t, the five evaluation times. Scene q contains K_q conditions and C_q action sequences, with $T = 5$ times. Let $\hat{z}_{qkct}$ and z_{qkct} denote predicted and true latent states, and let $\bar{z}_{qct} = K_q^{-1} \sum_k z_{qkct}$ be the mean true future across conditions for the same scene, actions, and time. The scene-level prediction error L_q and mean-future reference error B_q are

$$L_q = \frac{1}{K_q C_q T} \sum_{k,c,t} \|\hat{z}_{qkct} - z_{qkct}\|^2,$$
$$B_q = \frac{1}{K_q C_q T} \sum_{k,c,t} \|z_{qkct} - \bar{z}_{qct}\|^2. \tag{7}$$

We report their aggregate ratio [25]:

$$E/B = \frac{\sum_q L_q}{\sum_q B_q}, \tag{8}$$

where $E = \sum_q L_q$ and $B = \sum_q B_q$ are the total prediction and reference errors, respectively, with $B > 0$. Lower is better: $E/B = 0$ means exact predictions, while $E/B = 1$ matches the mean-future reference error. For equally weighted conditions sharing the current observation and future actions, this reference minimizes squared error among predictions that ignore the dynamics condition [18]; thus a deterministic history-independent model cannot achieve $E/B < 1$.

For the history diagnostic, we also report normalized history gain G_H. Let $\hat{z}_{qhct}$ be the prediction using history from condition h, while z_{qkct} is the true future under condition k, with the same current observation and future actions. The

aggregate matching- and mismatched-history errors are

$$E_+ = \sum_q \frac{1}{K_q C_q T} \sum_{k,c,t} \|\hat{z}_{qkct} - z_{qkct}\|^2,$$
$$E_- = \sum_q \frac{1}{K_q C_q T} \sum_{k,c,t} \frac{1}{K_q - 1} \sum_{h \neq k} \|\hat{z}_{qhct} - z_{qkct}\|^2. \tag{9}$$

The mismatched error averages equally over all $K_q - 1$ alternative histories. We compute

$$G_H = \frac{E_- - E_+}{B}, \tag{10}$$

using the same reference error B as above. A positive G_H means matching history reduces prediction error; unlike History win rate, it measures the size of this reduction and does not imply accurate prediction.

Evaluation Scope and Reporting. Adaptation Accuracy, History win rate, and NRE use Test for the main benchmark, as do multi-step prediction and additional mechanism analyses; training-query analyses use Training. The main task queries and extended multi-step evaluations differ in future actions and prediction horizons. Latent errors are measured in each checkpoint's own target representation and do not provide a common physical error scale across models. We report means and standard deviations across available training runs. For multi-step prediction, source-cluster bootstrap intervals reflect sampling uncertainty for the evaluated checkpoints, not training-run variability (Appendix E). We separately report CEM planning success in the original environments to assess retention of existing control capabilities (Section 4).

4. Experiments

4.1. Experimental Setup

Evaluated models and representation strategies. We evaluate three action-conditioned latent world models with different representation-learning strategies: (1) DINO-WM [37], which trains a dynamics predictor on frozen, pretrained DINOv2 features; (2) PLDM [30], which jointly optimizes the visual encoder and dynamics predictor; and (3) LeWM [24], which combines latent prediction with representation regularization to mitigate representation collapse. All model weights remain fixed during evaluation. Because these models also differ in architecture and training objectives, their performance differences cannot be attributed to representation-learning strategies alone.

Training schemes and design rationale. We evaluate four training schemes to examine how training choices affect in-context dynamics adaptation (Table 2).

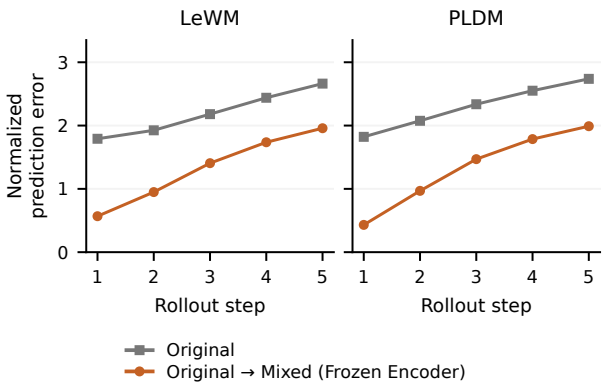


Figure 2. **Multi-step prediction on pusher displacement (Test).** Each rollout step spans five simulator steps without new observations. Prediction error is divided by the error of predicting the mean future representation across dynamics conditions. Ratios are computed within each checkpoint's representation, then averaged over runs. These extended queries differ from the one-step tasks; additional results appear in Appendix H.

Pretraining and data composition: The Original scheme uses only fixed-dynamics data to assess adaptation without exposure to ContextWorld training data. The Mixed dataset combines original-environment and ContextWorld training windows at a $1:1$ sampling ratio. We compare training on Mixed from initialization with training on Mixed after an Original training stage. The latter includes an additional 10 epochs of Original training, so the two schemes do not have equal total training budgets.

Visual encoder updates: For LeWM and PLDM, we compare two fine-tuning schemes starting from the same Original checkpoint. Joint updates the visual encoder and the remaining trainable modules, whereas Frozen Encoder fixes the visual encoder while training the remaining modules. This comparison examines the effect of freezing the visual encoder on adaptation, but does not freeze the entire latent representation, as the projector remains trainable.

Models are trained independently for each task. We evaluate planning success in the original environments using the Cross-Entropy Method (CEM) [12] to assess whether existing control performance is maintained. Optimization details are provided in Appendix G.

4.2. Adaptation Results

Original models trained under fixed dynamics have future-selection accuracy near 50% on most of the seven binary tasks (Table 3). Adding ContextWorld training data improves several tasks, including arm mass for LeWM and PLDM, but the gains are not uniform.

Under Mixed training, LeWM and PLDM reach 97.70% and 100.00% on action delay, whereas DINO-WM reaches

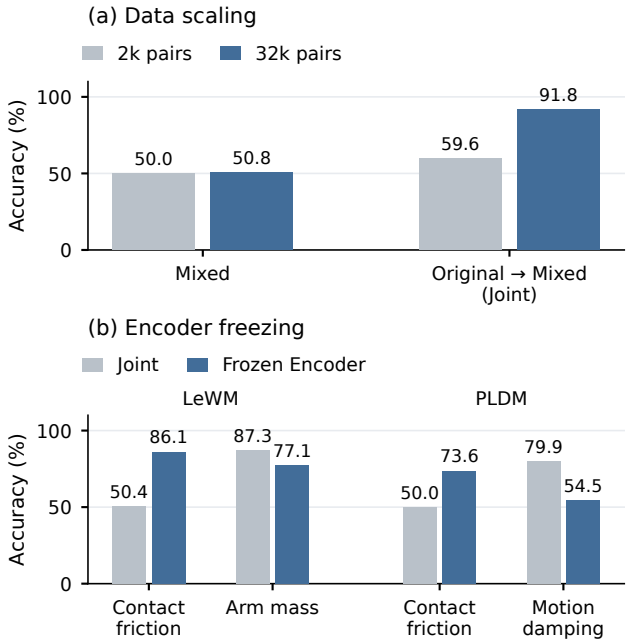


Figure 3. **Data scaling and encoder freezing on Test.** (a) LeWM on pusher displacement with 2,048 versus 32,768 ContextWorld training pairs; original-environment data are unchanged. (b) Joint versus frozen-encoder fine-tuning for LeWM and PLDM. Each comparison uses the same Original checkpoint and task data. Bars show accuracy (%). Training epochs are fixed. Full results and run counts appear in Table 6 and Appendix H.1.

16.65%. On pusher displacement, DINO-WM reaches 97.66%, outperforming both LeWM and PLDM under Mixed. All three models score at least 99.83% on door traversability. No model leads across all tasks.

On pusher displacement, Original → Mixed (Frozen Encoder) achieves Test History win rates of 98.83% for LeWM and 99.80% for PLDM, with NRE of 0.128 and 0.101. Yet their extended multi-step Test evaluations yield aggregate E/B of 1.436 and 1.448, respectively (Figure 2). Thus, strong task-query adaptation does not guarantee low complete prediction error under different actions and longer horizons; these evaluations do not isolate the cause of the errors.

4.3. Effects of Training Choices

Freezing the pretrained visual encoder helps some tasks and hurts others. For LeWM, it improves contact friction Test accuracy from 50.39% to 86.13% relative to joint training, but reduces arm mass accuracy from 87.30% to 77.15% (Figure 3b; Table 3). Each comparison holds the data, initialization, and training duration fixed. Because projectors and action encoders remain trainable, this intervention does not isolate a static target representation.

Figure 3a examines the effect of data scaling on Test accuracy. For LeWM on pusher displacement, data scaling substantially benefits Original → Mixed (Joint), whereas Mixed stays near chance. NRE also improves with joint fine-tuning (Appendix H.1). Both data scales use the same number of training updates.

Under the tested 10-epoch recipe, increasing LeWM and PLDM from 18M to 183M parameters yields no consistent Test accuracy gain on friction and damping (Appendix H.2).

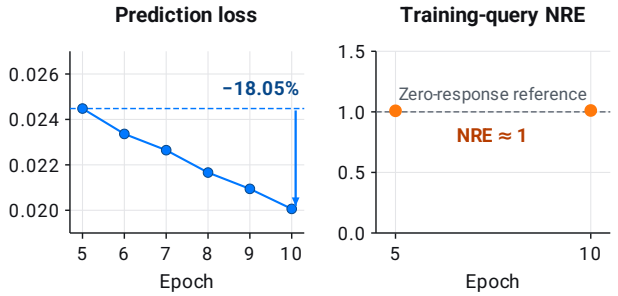


Figure 4. **Lower prediction loss does not ensure an accurate conditional response.** PLDM on contact friction, Original → Mixed (Joint). Recorded prediction loss falls by 18.05% from epochs 5 to 10, while training-query NRE stays near the zero-response reference of 1. Loss uses mixed training batches; NRE is measured on fixed paired training queries at epochs 5 and 10. Their adaptation accuracy is 49.51% and 49.71%, respectively. Full scheme and task comparisons appear in Figure 5.

4.4. Prediction Loss and Adaptation

Weak evaluation performance may reflect poor generalization or conditional responses that were not learned well during training. We compare prediction loss with NRE on fixed training queries at epochs 5 and 10 (Figure 4).

On friction, PLDM joint training reduces prediction loss by 18.05%, yet training-query NRE remains near the zero-response reference. On damping with the encoder frozen, loss approaches convergence while NRE improves but remains high (Figure 5). Changing the scheme helps: freezing yields a stronger conditional response on friction, whereas joint training performs better on damping, with Test showing the same task-dependent ordering.

These results expose a disconnect between prediction-loss optimization and history-conditioned adaptation, even on training data. Prediction loss aggregates errors across mixed data, whereas NRE directly measures error in the predicted differences between conditions. Optimizing the former does not ensure an accurate conditional response. This finding identifies insufficient constraints on history-conditioned dynamics in the evaluated training objectives and strategies.

5. Conclusion

ContextWorld establishes a standardized benchmark to evaluate in-context dynamics adaptation in action-conditioned latent visual world models under strictly frozen weights. Across nine tasks spanning four simulated environments, our paired history comparisons decouple whether models condition their predictions on interaction history from how accurately they forecast the resulting futures. Systematic evaluations of LeWM, PLDM, and DINO-WM demonstrate that while training on multi-dynamics data can improve adaptation, relative model advantages and the impact of visual encoder freezing remain task-dependent. Crucially, our training-distribution diagnostics expose a disconnect between prediction-loss optimization and in-context adaptation: conditional responses can remain poorly captured even as training loss steadily decreases or approaches convergence. These findings identify insufficient constraints on history-conditioned dynamics within the evaluated training objectives and strategies, underscoring the need for direct adaptation benchmarks and formulations that explicitly constrain history-conditioned responses.

Our findings are framed by specific diagnostic boundaries. Predictions are evaluated in each model's native representation rather than a shared physical error space; most tasks examine adaptation among dynamics conditions represented in training, while Speed evaluates unseen parameter values; and mixed-data checkpoints are trained independently per task, with planning experiments quantifying capability retention in unperturbed environments. These boundaries outline clear avenues for future research: establishing complementary cross-model evaluation in grounded physical coordinates, scaling toward generalist multi-task world models that generalize across broader dynamics, and examining closed-loop planning to test whether history-conditioned latent predictions translate directly into improved decision-making.

References

[1] Ossama Ahmed, Frederik Träuble, Anirudh Goyal, Alexander Neitz, Yoshua Bengio, Bernhard Schölkopf, Manuel Wüthrich, and Stefan Bauer. CausalWorld: A Robotic Manipulation Benchmark for Causal Structure and Transfer Learning. In *International Conference on Learning Representations*, 2021. 3

[2] Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-Supervised Learning From Images With a Joint-Embedding Predictive Architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15619–15629, 2023. 3

[3] Mahmoud Assran, Adrien Bardes, David Fan, Quentin Garrido, Russell Howes, Mojtaba Komeili, Matthew Muckley, Ammar Rizvi, Claire Roberts, Koustuv Sinha, Artem Zholus, Sergio Arnaud, Abha Gejji, Ada Martin, Francois Robert Hogan, Daniel Dugas, Piotr Bojanowski, Vasil Khalidov, Patrick Labatut, Francisco Massa, Marc Szafraniec, Kapil Krishnakumar, Yong Li, Xiaodong Ma, Sarath Chandar, Franziska Meier, Yann LeCun, Michael Rabbat, and Nicolas Ballas. V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. *arXiv preprint arXiv:2506.09985*, 2025. 3

[4] Fabien Baradel, Natalia Neverova, Julien Mille, Greg Mori, and Christian Wolf. CoPhy: Counterfactual Learning of Physical Dynamics. In *International Conference on Learning Representations*, 2020. 1, 3

[5] Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mahmoud Assran, and Nicolas Ballas. Revisiting Feature Prediction for Learning Visual Representations from Video. *arXiv preprint arXiv:2404.08471*, 2024. 3

[6] Jacob Beck, Risto Vuorio, Evan Zheran Liu, Zheng Xiong, Luisa Zintgraf, Chelsea Finn, and Shimon Whiteson. A Tutorial on Meta-Reinforcement Learning. *Foundations and Trends in Machine Learning*, 18(2–3):224–384, 2025. 3

[7] Carolin Benjamins, Theresa Eimer, Frederik Schubert, Aditya Mohan, Sebastian Döhler, André Biedenkapp, Bodo Rosenhahn, Frank Hutter, and Marius Lindauer. Contextualize Me – The Case for Context in Reinforcement Learning. *Transactions on Machine Learning Research*, 2023. 3

[8] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D. Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel Ziegler, Jeffrey Wu, Clemens Winter, Chris Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language Models are Few-Shot Learners. In *Advances in Neural Information Processing Systems*, 2020. 3

[9] Zhenfang Chen, Kexin Yi, Yunzhu Li, Mingyu Ding, Antonio Torralba, Joshua B. Tenenbaum, and Chuang Gan. ComPhy: Compositional Physical Reasoning of Objects and Events from Videos. In *International Conference on Learning Representations*, 2022. 3

[10] Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin CM Burchfiel, and Shuran Song. Diffusion Policy: Visuomotor Policy Learning via Action Diffusion. In *Proceedings of Robotics: Science and Systems*, 2023. 4

[11] Karl Cobbe, Chris Hesse, Jacob Hilton, and John Schulman. Leveraging Procedural Generation to Benchmark Reinforcement Learning. In *Proceedings of the 37th International Conference on Machine Learning*, pages 2048–2056. PMLR, 2020. 3

[12] Pieter-Tjerk de Boer, Dirk P. Kroese, Shie Mannor, and Reuven Y. Rubinstein. A Tutorial on the Cross-Entropy Method. *Annals of Operations Research*, 134(1):19–67, 2005. 7

[13] Qingxiu Dong, Lei Li, Damai Dai, Ce Zheng, Jingyuan Ma, Rui Li, Heming Xia, Jingjing Xu, Zhiyong Wu, Baobao Chang, Xu Sun, Lei Li, and Zhifang Sui. A Survey on In-context Learning. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 1107–1128. Association for Computational Linguistics, 2024. 3

[14] Yan Duan, John Schulman, Xi Chen, Peter L. Bartlett, Ilya Sutskever, and Pieter Abbeel. RL^2: Fast Reinforcement Learning via Slow Reinforcement Learning. *arXiv preprint arXiv:1611.02779*, 2016. 3

[15] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, 1993. 13

[16] Fabio Ferreira, Moreno Schlageter, Raghu Rajan, André Biedenkapp, and Frank Hutter. One-shot World Models Using a Transformer Trained on a Synthetic Prior. *arXiv preprint arXiv:2409.14084*, 2024. 3

[17] C. A. Field and A. H. Welsh. Bootstrapping Clustered Data. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 69(3):369–390, 2007. 13

[18] Tilmann Gneiting. Making and Evaluating Point Forecasts. *Journal of the American Statistical Association*, 106(494): 746–762, 2011. 6

[19] David Ha and Jürgen Schmidhuber. Recurrent World Models Facilitate Policy Evolution. In *Advances in Neural Information Processing Systems*, 2018. 1, 3

[20] Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning Latent Dynamics for Planning from Pixels. In *Proceedings of the 36th International Conference on Machine Learning*, pages 2555–2565. PMLR, 2019. 1, 3

[21] Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to Control: Learning Behaviors by Latent Imagination. In *International Conference on Learning Representations*, 2020. 1, 3

[22] Steeven Janny, Fabien Baradel, Natalia Neverova, Madiha Nadri, Greg Mori, and Christian Wolf. Filtered-CoPhy: Unsupervised Learning of Counterfactual Physics in Pixel Space. In *International Conference on Learning Representations*, 2022. 3

[23] Yann LeCun. A Path Towards Autonomous Machine Intelligence, 2022. Position paper. 3

[24] Lucas Maes, Quentin Le Lidec, Damien Scieur, Yann LeCun, and Randall Balestriero. LeWorldModel: Stable End-to-End Joint-Embedding Predictive Architecture from Pixels. *arXiv preprint arXiv:2603.19312*, 2026. 2, 3, 7

[25] Allan H. Murphy. Skill Scores Based on the Mean Square Error and Their Relationships to the Correlation Coefficient. *Monthly Weather Review*, 116(12):2417–2424, 1988. 6

[26] Alexander Nikulin, Vladislav Kurenkov, Ilya Zisman, Artem Agarkov, Viacheslav Sinii, and Sergey Kolesnikov. XLand-MiniGrid: Scalable Meta-Reinforcement Learning Environments in JAX. In *Advances in Neural Information Processing Systems*, 2024. 3

[27] Alexander Nikulin, Ilya Zisman, Alexey Zemtsov, and Vladislav Kurenkov. XLand-100B: A Large-Scale Multi-Task Dataset for In-Context Reinforcement Learning. In *International Conference on Learning Representations*, 2025. 3

[28] Seohong Park, Kevin Frans, Benjamin Eysenbach, and Sergey Levine. OGBench: Benchmarking Offline Goal-Conditioned RL. In *International Conference on Learning Representations*, 2025. 4

[29] Kate Rakelly, Aurick Zhou, Chelsea Finn, Sergey Levine, and Deirdre Quillen. Efficient Off-Policy Meta-Reinforcement Learning via Probabilistic Context Variables. In *Proceedings of the 36th International Conference on Machine Learning*, pages 5331–5340. PMLR, 2019. 3

[30] Vlad Sobal, Wancong Zhang, Kyunghyun Cho, Randall Balestriero, Tim G. J. Rudner, and Yann LeCun. Learning from Reward-Free Offline Data: A Case for Planning with Latent Dynamics Models. *arXiv preprint arXiv:2502.14819*, 2025. 2, 3, 7

[31] Yuval Tassa, Yotam Doron, Alistair Muldal, Tom Erez, Yazhe Li, Diego de Las Casas, David Budden, Abbas Abdolmaleki, Josh Merel, Andrew Lefrancq, Timothy Lillicrap, and Martin Riedmiller. DeepMind Control Suite. *arXiv preprint arXiv:1801.00690*, 2018. 4

[32] Hsiao-Yu Tung, Mingyu Ding, Zhenfang Chen, Daniel M. Bear, Chuang Gan, Joshua B. Tenenbaum, Daniel L. K. Yamins, Judith Fan, and Kevin A. Smith. Physion++: Evaluating Physical Scene Understanding that Requires Online Inference of Different Physical Properties. In *Advances in Neural Information Processing Systems*, 2023. 1, 3

[33] Fan Wang, Zhiyuan Chen, Yuxuan Zhong, Sunjian Zheng, Pengtao Shao, Bo Yu, Shaoshan Liu, Jianan Wang, Ning Ding, Yang Cao, and Yu Kang. Context and Diversity Matter: The Emergence of In-Context Learning in World Models. In *International Conference on Learning Representations*, 2026. 1, 3, 12

[34] Siyin Wang, Junhao Shi, Senyu Fei, Zhaoyang Fu, Li Ji, Jingjing Gong, and Xipeng Qiu. In-Context World Modeling for Robotic Control. *arXiv preprint arXiv:2606.26025*, 2026. 3

[35] Tianhe Yu, Deirdre Quillen, Zhanpeng He, Ryan Julian, Karol Hausman, Chelsea Finn, and Sergey Levine. Meta-World: A Benchmark and Evaluation for Multi-Task and Meta Reinforcement Learning. In *Proceedings of the Conference on Robot Learning*, pages 1094–1100. PMLR, 2020. 3

[36] Haochen Zhang, Jiaheng Guo, Zhen Xu, Zachary Plotkin, Nicholas Konz, Zhen Tan, and Tianlong Chen. On the Divergence of Accuracy and Mechanism Consistency in Time Series World Models. *arXiv preprint arXiv:2610.01842*, 2026. 3

[37] Gaoyue Zhou, Hengkai Pan, Yann LeCun, and Lerrel Pinto. DINO-WM: World Models on Pre-trained Visual Features enable Zero-shot Planning. In *Proceedings of the 42nd International Conference on Machine Learning*, pages 79115–79135. PMLR, 2025. 2, 3, 4, 7

A. Input Sampling

Observations are typically sampled every five simulator steps. The actions executed between consecutive observations are included in the interaction history.

B. Trajectory Construction and Pairing

Continuous trajectories. Each sample follows a continuous simulator trajectory from its history through its future, without intermediate resets or state overrides. Identical current observations may still conceal differences in velocity or contact state, so we perform task-specific checks of physical-state matching at the start of prediction. For action delay, we retain the actual queue of pending actions.

Starting positions. For tasks in which the two dynamics conditions cannot naturally share a starting position, we exchange their starting positions during generation and verify that the two conditions have identical sets of starting images. This check reduces the possibility that the starting image alone reveals the condition.

Predictions without history. Consider a complete, balanced pair with distinguishable encoded targets. A deterministic model that ignores history receives the same current observation and future action sequence for both samples and therefore produces the same prediction. This prediction cannot be strictly closer to each sample's own target than to the other target. Consequently, such a model scores at most 50% on the paired future-selection metric in Eq. 3; ties count as failures.

C. Action Delay Response Groups

The delay data retain all 11 delays from 0 to 10 simulator steps. At the five-step horizon used for the main one-step score, delays of 5–10 produce the same stationary response and are grouped together. This score therefore uses six distinguishable response groups: 0, 1, 2, 3, 4, and 5–10.

D. Dataset Statistics and Splits

D.1. Dataset Sizes

Table 4 lists the task data used in the training and evaluation comparisons. Development supports method and checkpoint selection; Test is reserved for final evaluation. Training sizes exclude the original-environment data used for mixed training. A pair contains two trajectories with different dynamics conditions; a trajectory count refers to recorded episodes, not to the number of training windows sampled from them. An evaluation case fixes the current observation and future action sequence and can include histories and targets from several dynamics conditions.

The action delay counts refer to the dataset covering all delays from 0 to 10 simulator steps. Its Training split contains 5,120 base cases, each simulated under all 11 delays, giving 56,320 trajectories. Development and Test each contain 300 base cases, again evaluated under all 11 delays. Door Training contains 7,680 trajectories for each of the passable and blocked rules.

Several examples can share a source episode: pusher displacement can use multiple action windows from one episode, and forward and mirrored damping examples can share a source. The 10,000 gripper carry training pairs use 10,000 distinct source episodes.

D.2. Paired Dynamics Conditions

Pusher displacement uses action gains of 60 and 140. Contact friction uses coefficients 0.05 and 0.80 between the pusher and the object. Motion damping fixes effective contact friction at 0.25 and uses simulator damping settings 0.2 and 1.0; here 0.2 gives faster decay, while 1.0 applies no additional decay. Arm mass uses densities of 500 and 1,500 for the MuJoCo arm link and finger geometry; the hand link retains its original density. Gripper carry changes whether the closed gripper carries the cube. Portal exit uses two exit rules for the same entrance, with paired trajectories reaching the shared query state before entering it. The numeric settings are simulator parameters, not additional model inputs.

D.3. Speed Values and Door Positions

Table 5 lists the speed values used for Training and evaluation. Development and Test use the same evaluation speeds but different interaction trajectories. Each split contains 300 cases for each of the 14 evaluated speeds, or 4,200 cases in total before comparing histories from different speeds. The main task results use the unseen speeds within the training range.

Door position is specified by an integer coordinate along the wall. Test uses the 42 positions $30 + 4k$, where $k = 0, \ldots, 41$. Development uses the 16 positions 65, 77, 79, 87, 112, 115, 131, 152, 167, 169, 171, 172, 175, 180, 183, 185. Training uses the integers from 36 through 188 after excluding all Development and Test positions and the additional unused positions 40, 127, and 132, leaving 96 positions. The three position sets are disjoint. Both passable and blocked rules are used across the splits; the test distinction is the door position, not a new rule.

D.4. Split Checks and Model Inputs

Split checks use source episode identifiers, trajectory template identifiers, or observation hashes, as appropriate for each task. Speed Development uses independently generated initial configurations and excludes initial states and current observations that duplicate Test cases. For Door, we

Table 4. Dataset sizes and condition coverage.

Task	Training	Development	Test
Speed	16,384 trajectories; 32 speeds	300 cases per speed	300 cases per speed
Pusher displacement	32,768 pairs	256 pairs	256 pairs
Arm mass	32,768 pairs	256 pairs	256 pairs
Action delay	56,320 trajectories; 11 delays	300 cases $\times$ 11 delays	300 cases $\times$ 11 delays
Contact friction	32,768 pairs	256 pairs	256 pairs
Motion damping	32,768 pairs	256 pairs	256 pairs
Gripper carry rule	10,000 pairs	256 pairs	256 pairs
Door traversability	15,360 trajectories; 96 door positions	300 cases; 16 door positions	300 cases; 42 door positions
Portal exit	32,768 pairs	256 pairs	256 pairs

Table 5. Speed values in the benchmark. The four evaluation groups are used in both Development and Test. Interpolation values lie within the Training range but are absent from Training.

Setting	Speed values
Training	2.6, 2.7, 2.8, 2.9, 3.1, 3.3, 3.5, 3.7, 3.8, 3.9, 4.1, 4.2, 4.4, 4.5, 4.7, 5.0, 5.1, 5.3, 5.5, 5.7, 5.9, 6.1, 6.2, 6.3, 6.6, 6.7, 6.8, 7.0, 7.2, 7.3, 7.8, 7.9
Seen speeds	3.1, 5.1, 7.0
Interpolation	3.4, 4.8, 6.9
Lower extrapolation	1.75, 1.95, 2.15, 2.35
Upper extrapolation	8.25, 8.75, 9.5, 10.25

also check the position sets listed above for overlap.

Hidden dynamics values, simulator states, and pair identifiers are retained for data generation, validation, and scoring. They are not passed to the prediction model, which receives only the visual observations and actions defined in Sec. 3.1.

E. Evaluation Details

Paired tasks. The strict comparisons in Eq. 3 do not count ties as correct. The two comparisons within a pair are dependent. Appendix B explains the upper bound for a deterministic model that ignores history.

Speed. For each target speed, accuracy is the proportion of cases in which matching history gives strictly lower prediction error than every alternative history. We then average accuracy equally across target speeds. Ties do not count as correct: if all histories produce the same prediction, accuracy is zero. The main results use the interpolation speeds listed in Appendix D.

Action delay. We select the candidate true future with the lowest mean squared error, resolving ties in favor of the smaller delay. The prediction is correct when the selected delay and the true delay belong to the same response group. Within each case, we first average correctness over delays in each group, then average equally across the six groups defined in Appendix C. The final accuracy is the mean over cases.

Door traversability. For a prediction made with matching history, we compare the mean squared errors to the passage and blockage futures. The prediction is correct only if the error to the true future is strictly smaller than the error to the alternative future. Accuracy is averaged over cases and the two rules; ties do not count as correct.

Multi-step prediction horizons. Each prediction step advances five simulator steps. We measure prediction errors at 5, 10, 15, 20, and 25 simulator steps after the current observation.

Within-scene aggregation. Let z_{qkct} and $\hat{z}_{qkct}$ denote the true and predicted future representations for scene q, dynamics condition k, action sequence c, and prediction time t. With K_q conditions, C_q shared action sequences, and $T = 5$ prediction times, the errors used in Eq. 8 are

$$L_q = \frac{1}{K_q C_q T} \sum_{k,c,t} \|\hat{z}_{qkct} - z_{qkct}\|^2,$$

$$B_q = \frac{1}{K_q C_q T} \sum_{k,c,t} \|z_{qkct} - \bar{z}_{qct}\|^2,$$

$$\bar{z}_{qct} = \frac{1}{K_q} \sum_{k} z_{qkct}.$$

L2World tests history dependence by shuffling context observations [33]; our controls instead swap histories between dynamics conditions. For history controls, E_+ and E_- sum the matching- and mismatched-history scene errors using the same averaging over conditions, action sequences, and

prediction times. When a sample has multiple mismatched histories, their errors are averaged equally. A positive G_H indicates that matching history reduces error; it does not imply a low complete prediction error.

Mean-future reference. The reference in Eq. 8 is used only for scoring and is never provided to the model. The ratio is defined only when $\sum_q B_q > 0$. When conditions share the current observation and future actions and are equally weighted, a deterministic model that ignores history cannot achieve $E/B < 1$: it must make the same prediction across conditions, and the mean-future reference minimizes the error of such a prediction.

Response error settings. We specify the prediction horizon for each NRE analysis. The ratio in Eq. 6 is defined only when $\sum_i \|\Delta z_i\|^2 > 0$. NRE equal to 1 does not by itself show that a model ignores history. Errors shared by both predictions cancel in the difference, so an exact conditional response does not imply exact complete predictions.

Uncertainty estimates. For multi-step errors, we use cluster bootstrap resampling [15, 17] at the level of source trajectories or scenes, retaining related samples together. These intervals describe scene-sampling uncertainty for the evaluated checkpoints, not variability across training runs.

Interpreting multiple metrics. The metrics describe different aspects of the same predictions and are not independent pieces of evidence.

F. Scoring Controls

We validate the paired scorer using constructed predictions on 256 Test pusher-displacement scenes, separately in the target representations of LeWM Original and Original → Mixed (Frozen Encoder). The queries use a nonzero action candidate and the five-step prediction horizon; the predictor is not run for this check. Both representations give the results below. With $m = (z_A + z_B)/2$, even predictions $m + 0.001(z_k - m)$ select the correct future while recovering only a tiny fraction of the true response. This control explains why we report response error alongside task accuracy. It verifies the scorer on these queries, not physical calibration across all representations.

Constructed prediction	Accuracy (%)	History win rate (%)	NRE
True target for each condition	100	100	0
Same target z_A for both conditions	50	0	1
Same midpoint m for both conditions	0	0	1
Swapped true targets	0	0	4
$m + 0.001(z_k - m)$ for condition k	100	100	0.998001

G. Training and Diagnostic Settings

Optimization. The training recipe specifies 10 epochs per stage on eight GPUs, with 128 windows per device and no gradient accumulation, giving a global batch size of 1,024. LeWM and PLDM use AdamW with learning rate 5×10^{-5} and weight decay 0.001. DINO-WM uses learning rate 5×10^{-4} and zero weight decay; its pretrained DINOv2-small visual encoder remains frozen. The models retain their respective prediction and regularization losses. The Original → Mixed schemes load model weights but reset optimizer state. Encoder freezing fixes both parameters and training-state buffers; it does not freeze the projector or action encoder.

Input and sampling. The standard setup uses three observation frames, five simulator steps between frames, and a one-frame prediction target. Action delay mixed training instead uses seven-frame history. The original-environment data are the TwoRoom, PushT, Reacher, and Cube datasets released with LeWM; each task uses only the dataset for its environment. Only the ContextWorld Training split is mixed into training. For component lengths N_{orig} and N_{CW}, the mixture has a nominal epoch length of $2\max(N_{\mathrm{orig}}, N_{\mathrm{CW}})$ windows before partitioning and batching, with equal counts from the two components. The smaller component is sampled repeatedly. Consequently, equal epoch counts do not guarantee equal update counts across data scales, and the two-stage schemes include additional original-environment training.

Prediction diagnostics. The main task comparison uses Test. The multi-step and history diagnostics below use Test and the same checkpoints as the main task comparison. Multi-step evaluation covers 2,436 source scenes across all nine tasks and predicts at 5, 10, 15, 20, and 25 simulator steps using only the initial observed history. The candidate actions and multi-step targets extend the one-step task queries. For gripper carry, we replay all 256 held-out source cases and render their histories and extended futures together, preserving the paired current-observation constraint without mixing stored images with newly rendered futures. These coherent replays retain the source conditions and actions but do not reproduce the stored images exactly; the original task-query evaluation is unchanged. Figure 2 reports E_t/B_t: prediction and mean-future errors are summed over queries separately at each horizon, normalized within each checkpoint, then averaged equally over training repetitions. The aggregate E/B instead combines errors over all horizons before division; it is not the arithmetic mean of these per-horizon ratios. Each checkpoint is scored separately, and matching training repetitions are then averaged equally. Intervals use 4,000 source-cluster bootstrap samples; all conditions, actions, and times from a source remain together. They condition on the evaluated checkpoints and do not estimate variation from retraining the model. Original action-delay checkpoints do not have the native seven-frame interface and are excluded from this comparison. DINO-WM has no jointly updated visual-encoder scheme; its comparable Original → Mixed results cover Speed, damping, and gripper carry.

Planning evaluation. CEM results measure success in the original environments, with no hidden-dynamics history intervention. Most checkpoint evaluations use six evaluation seeds with 50 episodes each; some original LeWM and PLDM references use three sets of 100 episodes or an aggregated 300-episode result. Evaluation repetitions are pooled within each checkpoint, then training repetitions are averaged equally. A single original-environment checkpoint supplies the reference for all tasks from that environment; these repeated entries are not independent experiments.

Training-query analysis. The friction and damping comparisons use 32,768 training pairs, identical original-environment initialization within each task, and the same 50/50 mixture, batch size, and duration. We evaluate 512 fixed training pairs and 256 Test pairs per task at epochs 5 and 10, predicting the next observation five simulator steps ahead. Loss summaries use the logged mixed-training batches, approximately 50–70 records per epoch, rather than every training batch. Checkpoints and logs are aligned by update step. The curves show recorded means and batch interquartile ranges; loss medians give the same monotonic direction for friction joint training from epochs 5 to 10. NRE is evaluated in each checkpoint's target representation and is not a common physical error across checkpoints. For damping with the encoder frozen, prediction loss changes by only -0.39% between these endpoints and fluctuates in between. The late-training plateau is consistent with the loss approaching convergence.

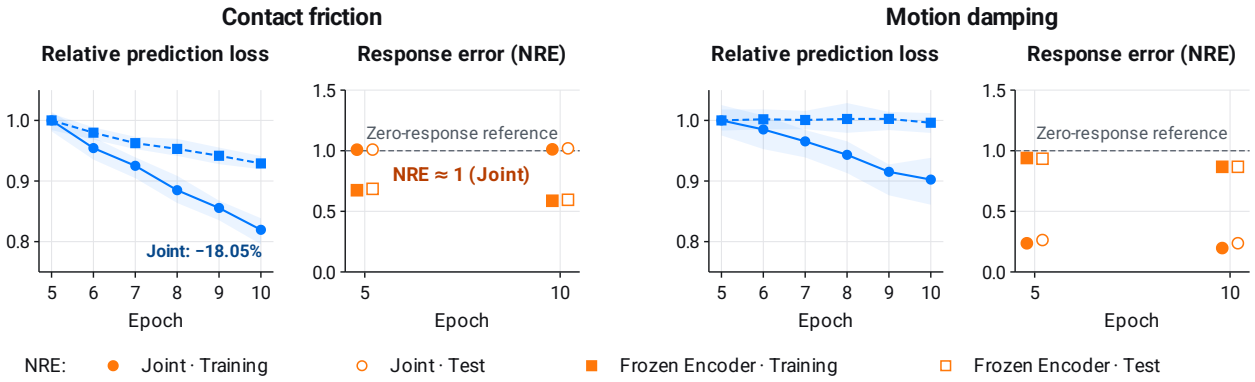


Figure 5. **Training loss and conditional responses across tasks and schemes.** Within each task, PLDM Joint and Frozen Encoder share Original initialization. Loss is normalized to epoch 5; shading shows batch interquartile ranges. Blue denotes prediction loss and orange denotes NRE; circles denote Joint and squares denote Frozen Encoder. NRE uses filled markers for Training and open markers for Test, evaluated only at epochs 5 and 10. Small horizontal offsets separate the two splits; 1 marks the zero-response reference. Loss uses mixed batches, whereas NRE uses fixed paired task queries.

H. Additional Results

Table 6 reports all three Test adaptation metrics with training-run counts and standard deviations. Table 7 combines Test history and response metrics with multi-step prediction errors and original-environment planning results. History win rate measures strict error reduction from matching history on the task queries; NRE measures the conditional response on those queries. E/B uses the extended multi-step action panels and includes errors shared by both conditions. Because the query sets and prediction horizons differ, these columns should not be treated as measurements of the same event. All latent errors are computed in the checkpoint's own target representation.

Table 6. **Complete Test adaptation metrics.** Each row uses the same evaluated checkpoints for all three metrics. Entries are means $\pm$ sample standard deviations across n training runs; a single run has no sample standard deviation. Accuracy and History win rate are percentages; NRE is dimensionless. Speed uses unseen interpolation speeds, and its Accuracy and History win rate use the same scoring rule. Action delay uses seven-frame history. Only available comparable configurations are listed.

Task	Model	Training scheme	n	Accuracy ↑	History win rate ↑	NRE ↓
Speed	LeWM	Original	3	58.04 ± 1.70	58.04 ± 1.70	0.896 ± 0.027
Speed	LeWM	Mixed	3	98.19 ± 0.57	98.19 ± 0.57	0.042 ± 0.011
Speed	LeWM	Original → Mixed (Joint)	1	96.89	96.89	0.055
Speed	LeWM	Original → Mixed (Frozen Encoder)	1	98.44	98.44	0.030
Speed	PLDM	Original	3	48.41 ± 4.33	48.41 ± 4.33	0.922 ± 0.026
Speed	PLDM	Mixed	3	96.78 ± 2.02	96.78 ± 2.02	0.084 ± 0.013
Speed	PLDM	Original → Mixed (Joint)	1	98.33	98.33	0.047

Task	Model	Training scheme	n	Accuracy ↑	History win rate ↑	NRE ↓
Speed	PLDM	Original → Mixed (Frozen Encoder)	1	96.44	96.44	0.062
Speed	DINO-WM	Original	1	57.11	57.11	0.919
Speed	DINO-WM	Mixed	3	59.41 ± 3.87	59.41 ± 3.87	0.961 ± 0.011
Speed	DINO-WM	Original → Mixed	1	84.78	84.78	0.411
Pusher displacement	LeWM	Original	3	48.96 ± 1.00	28.32 ± 12.76	1.018 ± 0.016
Pusher displacement	LeWM	Mixed	1	50.78	79.10	0.974
Pusher displacement	LeWM	Original → Mixed (Joint)	1	91.80	99.02	0.215
Pusher displacement	LeWM	Original → Mixed (Frozen Encoder)	1	97.46	98.83	0.128
Pusher displacement	PLDM	Original	3	45.57 ± 1.88	9.38 ± 4.20	1.058 ± 0.012
Pusher displacement	PLDM	Mixed	1	56.25	68.16	0.835
Pusher displacement	PLDM	Original → Mixed (Joint)	1	59.18	73.44	0.740
Pusher displacement	PLDM	Original → Mixed (Frozen Encoder)	1	98.63	99.80	0.101
Pusher displacement	DINO-WM	Original	1	38.48	9.38	1.127
Pusher displacement	DINO-WM	Mixed	1	97.66	99.61	0.394
Arm mass	LeWM	Original	3	50.07 ± 0.11	51.63 ± 0.60	0.977 ± 0.002
Arm mass	LeWM	Mixed	1	85.74	89.84	0.045
Arm mass	LeWM	Original → Mixed (Joint)	1	87.30	93.36	0.038
Arm mass	LeWM	Original → Mixed (Frozen Encoder)	1	77.15	86.52	0.346
Arm mass	PLDM	Original	3	51.82 ± 1.30	66.99 ± 9.68	0.921 ± 0.016
Arm mass	PLDM	Mixed	1	91.02	93.55	0.031
Arm mass	PLDM	Original → Mixed (Joint)	1	84.77	90.82	0.073
Arm mass	PLDM	Original → Mixed (Frozen Encoder)	1	81.84	89.26	0.213
Arm mass	DINO-WM	Original	1	50.78	61.33	0.977
Arm mass	DINO-WM	Mixed	1	78.32	88.09	0.673
Action delay	LeWM	Mixed	3	97.70 ± 0.59	99.18 ± 0.38	0.013 ± 0.002
Action delay	PLDM	Mixed	3	100.00 ± 0.00	100.00 ± 0.00	0.006 ± 0.001
Action delay	DINO-WM	Mixed	3	16.65 ± 0.03	19.45 ± 0.11	1.000 ± 0.000
Contact friction	LeWM	Original	3	49.93 ± 0.23	50.13 ± 0.63	1.001 ± 0.004
Contact friction	LeWM	Mixed	1	50.00	49.41	1.000
Contact friction	LeWM	Original → Mixed (Joint)	1	50.39	54.30	0.999
Contact friction	LeWM	Original → Mixed (Frozen Encoder)	1	86.13	92.77	0.429
Contact friction	PLDM	Original	3	50.00 ± 0.34	49.09 ± 1.58	1.014 ± 0.010
Contact friction	PLDM	Mixed	1	50.00	48.63	1.001
Contact friction	PLDM	Original → Mixed (Joint)	1	50.00	46.48	1.020
Contact friction	PLDM	Original → Mixed (Frozen Encoder)	1	73.63	83.59	0.595
Contact friction	DINO-WM	Original	1	50.39	47.66	1.015
Contact friction	DINO-WM	Mixed	1	84.18	92.19	0.648

Task	Model	Training scheme	n	Accuracy ↑	History win rate ↑	NRE ↓
Motion damping	LeWM	Original	3	50.00 ± 0.00	48.63 ± 1.60	1.007 ± 0.009
Motion damping	LeWM	Mixed	1	50.00	48.24	1.004
Motion damping	LeWM	Original → Mixed (Joint)	1	50.00	49.61	1.001
Motion damping	LeWM	Original → Mixed (Frozen Encoder)	1	53.91	61.91	0.958
Motion damping	PLDM	Original	3	50.26 ± 0.30	50.52 ± 0.63	0.947 ± 0.007
Motion damping	PLDM	Mixed	1	58.20	55.86	2.435
Motion damping	PLDM	Original → Mixed (Joint)	1	79.88	81.84	0.237
Motion damping	PLDM	Original → Mixed (Frozen Encoder)	1	54.49	64.26	0.867
Motion damping	DINO-WM	Original	1	49.80	49.61	1.014
Motion damping	DINO-WM	Mixed	1	56.45	65.23	1.042
Motion damping	DINO-WM	Original → Mixed	1	55.27	61.13	1.111
Gripper carry rule	LeWM	Original	3	52.21 ± 1.11	73.76 ± 1.97	0.909 ± 0.029
Gripper carry rule	LeWM	Mixed	1	50.00	49.80	1.000
Gripper carry rule	LeWM	Original → Mixed (Joint)	1	50.00	52.34	0.997
Gripper carry rule	LeWM	Original → Mixed (Frozen Encoder)	1	76.37	84.57	0.271
Gripper carry rule	PLDM	Original	3	50.78 ± 0.20	57.42 ± 1.35	0.937 ± 0.007
Gripper carry rule	PLDM	Mixed	1	49.80	50.78	0.997
Gripper carry rule	PLDM	Original → Mixed (Joint)	1	50.20	54.30	0.982
Gripper carry rule	PLDM	Original → Mixed (Frozen Encoder)	1	52.93	57.81	0.606
Gripper carry rule	DINO-WM	Original	1	50.00	50.59	1.000
Gripper carry rule	DINO-WM	Mixed	1	60.16	77.34	0.892
Gripper carry rule	DINO-WM	Original → Mixed	1	62.89	80.27	0.871
Door traversability	LeWM	Original	3	50.33 ± 0.00	50.94 ± 2.84	0.996 ± 0.003
Door traversability	LeWM	Mixed	3	99.83 ± 0.29	100.00 ± 0.00	0.041 ± 0.018
Door traversability	LeWM	Original → Mixed (Joint)	1	99.83	100.00	0.038
Door traversability	LeWM	Original → Mixed (Frozen Encoder)	1	100.00	100.00	0.036
Door traversability	PLDM	Original	3	50.06 ± 0.10	51.50 ± 0.67	0.997 ± 0.002
Door traversability	PLDM	Mixed	3	100.00 ± 0.00	100.00 ± 0.00	0.022 ± 0.005
Door traversability	PLDM	Original → Mixed (Joint)	1	100.00	100.00	0.023
Door traversability	PLDM	Original → Mixed (Frozen Encoder)	1	100.00	100.00	0.048
Door traversability	DINO-WM	Original	1	50.00	64.33	0.985
Door traversability	DINO-WM	Mixed	3	100.00 ± 0.00	100.00 ± 0.00	0.217 ± 0.003
Portal exit	LeWM	Original	3	50.07 ± 0.11	51.95 ± 0.70	0.990 ± 0.003
Portal exit	LeWM	Mixed	1	50.78	76.37	0.922
Portal exit	LeWM	Original → Mixed (Joint)	1	50.39	66.80	0.934
Portal exit	LeWM	Original → Mixed (Frozen Encoder)	1	77.93	94.34	0.264
Portal exit	PLDM	Original	3	50.07 ± 0.11	51.50 ± 0.30	0.996 ± 0.001
Portal exit	PLDM	Mixed	1	72.27	79.10	0.128
Portal exit	PLDM	Original → Mixed (Joint)	1	52.54	61.52	0.708
Portal exit	PLDM	Original → Mixed (Frozen Encoder)	1	91.21	98.83	0.191
Portal exit	DINO-WM	Original	1	50.78	56.45	0.996
Portal exit	DINO-WM	Mixed	1	100.00	100.00	0.142

Table 7. Test prediction diagnostics and original-environment CEM. Entries average the available training repetitions; E/B intervals are 95% source-bootstrap intervals. History win rate and CEM are percentages. Scheme names match the main text.

Task	Model	Scheme	History win rate ↑	NRE ↓	E/B ↓	E/B interval	CEM ↑
Speed	LeWM	Original	58.04	0.896	1.512	[1.491, 1.533]	92.44
Speed	LeWM	Mixed	98.19	0.042	0.368	[0.350, 0.388]	97.44
Speed	LeWM	Original → Mixed (Joint)	96.89	0.055	0.399	[0.376, 0.423]	97.00
Speed	LeWM	Original → Mixed (Frozen Encoder)	98.44	0.030	0.145	[0.133, 0.157]	95.00
Speed	PLDM	Original	48.41	0.922	3.446	[3.366, 3.542]	90.56
Speed	PLDM	Mixed	96.78	0.084	0.515	[0.492, 0.539]	96.89
Speed	PLDM	Original → Mixed (Joint)	98.33	0.047	0.380	[0.355, 0.408]	96.33
Speed	PLDM	Original → Mixed (Frozen Encoder)	96.44	0.062	0.308	[0.290, 0.326]	98.00
Speed	DINO-WM	Original	57.11	0.919	1.967	[1.949, 1.987]	100.00
Speed	DINO-WM	Mixed	59.41	0.961	2.086	[2.054, 2.123]	99.89
Speed	DINO-WM	Original → Mixed	84.78	0.411	0.886	[0.851, 0.924]	100.00
Pusher displacement	LeWM	Original	28.32	1.018	2.279	[2.211, 2.354]	82.22
Pusher displacement	LeWM	Mixed	79.10	0.974	3.779	[3.596, 3.968]	78.67
Pusher displacement	LeWM	Original → Mixed (Joint)	99.02	0.215	2.426	[2.353, 2.504]	86.00
Pusher displacement	LeWM	Original → Mixed (Frozen Encoder)	98.83	0.128	1.436	[1.365, 1.510]	73.67
Pusher displacement	PLDM	Original	9.38	1.058	2.386	[2.332, 2.442]	75.67
Pusher displacement	PLDM	Mixed	68.16	0.835	6.284	[6.001, 6.589]	61.33
Pusher displacement	PLDM	Original → Mixed (Joint)	73.44	0.740	3.720	[3.604, 3.840]	72.00
Pusher displacement	PLDM	Original → Mixed (Frozen Encoder)	99.80	0.101	1.448	[1.397, 1.501]	59.33
Pusher displacement	DINO-WM	Original	9.38	1.127	3.140	[3.024, 3.261]	76.67
Pusher displacement	DINO-WM	Mixed	99.61	0.394	1.931	[1.884, 1.981]	76.67
Arm mass	LeWM	Original	51.63	0.977	741.737	[644.066, 844.093]	55.89
Arm mass	LeWM	Mixed	89.84	0.045	51.802	[47.895, 56.217]	58.00
Arm mass	LeWM	Original → Mixed (Joint)	93.36	0.038	47.590	[44.059, 51.609]	59.00
Arm mass	LeWM	Original → Mixed (Frozen Encoder)	86.52	0.346	33.070	[30.472, 35.587]	59.67
Arm mass	PLDM	Original	66.99	0.921	19.698	[17.799, 21.952]	69.67
Arm mass	PLDM	Mixed	93.55	0.031	13.815	[12.861, 14.967]	51.67
Arm mass	PLDM	Original → Mixed (Joint)	90.82	0.073	46.000	[42.230, 50.596]	58.67
Arm mass	PLDM	Original → Mixed (Frozen Encoder)	89.26	0.213	17.048	[16.034, 18.176]	89.00
Arm mass	DINO-WM	Original	61.33	0.977	49.483	[46.626, 52.747]	90.67
Arm mass	DINO-WM	Mixed	88.09	0.673	14.166	[13.764, 14.537]	75.00
Action delay	LeWM	Mixed	99.18	0.013	1.429	[1.406, 1.453]	87.33

Task	Model	Scheme	History win rate ↑	NRE ↓	E/B ↓	E/B interval	CEM ↑
Action delay	PLDM	Mixed	100.00	0.006	1.818	[1.804, 1.832]	90.89
Action delay	DINO-WM	Mixed	19.45	1.000	66.094	[65.265, 66.948]	33.56
Contact friction	LeWM	Original	50.13	1.001	2.716	[2.642, 2.791]	82.22
Contact friction	LeWM	Mixed	49.41	1.000	2.727	[2.678, 2.781]	76.67
Contact friction	LeWM	Original → Mixed (Joint)	54.30	0.999	2.717	[2.677, 2.757]	86.00
Contact friction	LeWM	Original → Mixed (Frozen Encoder)	92.77	0.429	2.902	[2.843, 2.960]	83.33
Contact friction	PLDM	Original	49.09	1.014	3.809	[3.686, 3.940]	75.67
Contact friction	PLDM	Mixed	48.63	1.001	6.392	[6.077, 6.741]	65.67
Contact friction	PLDM	Original → Mixed (Joint)	46.48	1.020	7.236	[6.961, 7.541]	73.33
Contact friction	PLDM	Original → Mixed (Frozen Encoder)	83.59	0.595	4.536	[4.408, 4.666]	79.67
Contact friction	DINO-WM	Original	47.66	1.015	6.252	[5.971, 6.531]	76.67
Contact friction	DINO-WM	Mixed	92.19	0.648	3.695	[3.611, 3.776]	75.67
Motion damping	LeWM	Original	48.63	1.007	5.341	[5.105, 5.611]	82.22
Motion damping	LeWM	Mixed	48.24	1.004	248.446	[216.917, 288.116]	78.00
Motion damping	LeWM	Original → Mixed (Joint)	49.61	1.001	87.139	[77.847, 98.646]	88.33
Motion damping	LeWM	Original → Mixed (Frozen Encoder)	61.91	0.958	8.316	[7.878, 8.794]	84.33
Motion damping	PLDM	Original	50.52	0.947	6.216	[5.937, 6.535]	75.67
Motion damping	PLDM	Mixed	55.86	2.435	21.427	[18.508, 25.342]	61.00
Motion damping	PLDM	Original → Mixed (Joint)	81.84	0.237	8.323	[8.060, 8.597]	70.00
Motion damping	PLDM	Original → Mixed (Frozen Encoder)	64.26	0.867	5.680	[5.431, 5.959]	78.33
Motion damping	DINO-WM	Original	49.61	1.014	8.442	[8.184, 8.713]	76.67
Motion damping	DINO-WM	Mixed	65.23	1.042	7.292	[7.117, 7.484]	76.67
Motion damping	DINO-WM	Original → Mixed	61.13	1.111	6.975	[6.800, 7.165]	76.00
Gripper carry rule	LeWM	Original	73.76	0.909	10.647	[9.807, 11.634]	65.44
Gripper carry rule	LeWM	Mixed	49.80	1.000	170.794	[132.250, 222.804]	61.33
Gripper carry rule	LeWM	Original → Mixed (Joint)	52.34	0.997	12.148	[10.850, 13.816]	71.00

Task	Model	Scheme	History win rate ↑	NRE ↓	E/B ↓	E/B interval	CEM ↑
Gripper carry rule	LeWM	Original → Mixed (Frozen Encoder)	84.57	0.271	11.149	[10.363, 12.008]	67.00
Gripper carry rule	PLDM	Original	57.42	0.937	11.994	[9.809, 15.328]	53.44
Gripper carry rule	PLDM	Mixed	50.78	0.997	138.687	[115.564, 166.941]	55.00
Gripper carry rule	PLDM	Original → Mixed (Joint)	54.30	0.982	26.192	[23.629, 29.169]	56.67
Gripper carry rule	PLDM	Original → Mixed (Frozen Encoder)	57.81	0.606	25.681	[20.138, 33.793]	56.67
Gripper carry rule	DINO-WM	Original	50.59	1.000	40.837	[39.662, 42.051]	72.67
Gripper carry rule	DINO-WM	Mixed	77.34	0.892	11.402	[11.088, 11.709]	71.67
Gripper carry rule	DINO-WM	Original → Mixed	80.27	0.871	11.286	[10.982, 11.585]	73.67
Door traversability	LeWM	Original	50.94	0.996	5.255	[5.151, 5.364]	92.44
Door traversability	LeWM	Mixed	100.00	0.041	3.027	[2.984, 3.070]	95.11
Door traversability	LeWM	Original → Mixed (Joint)	100.00	0.038	2.309	[2.261, 2.358]	97.67
Door traversability	LeWM	Original → Mixed (Frozen Encoder)	100.00	0.036	2.495	[2.459, 2.528]	94.00
Door traversability	PLDM	Original	51.50	0.997	8.278	[8.069, 8.487]	90.56
Door traversability	PLDM	Mixed	100.00	0.022	2.370	[2.352, 2.389]	95.78
Door traversability	PLDM	Original → Mixed (Joint)	100.00	0.023	2.352	[2.318, 2.387]	99.33
Door traversability	PLDM	Original → Mixed (Frozen Encoder)	100.00	0.048	2.449	[2.417, 2.479]	98.67
Door traversability	DINO-WM	Original	64.33	0.985	3.812	[3.710, 3.915]	100.00
Door traversability	DINO-WM	Mixed	100.00	0.217	1.261	[1.238, 1.286]	97.56
Portal exit	LeWM	Original	51.95	0.990	908.548	[650.682, 1395.382]	92.44
Portal exit	LeWM	Mixed	76.37	0.922	47.098	[35.158, 68.065]	98.33
Portal exit	LeWM	Original → Mixed (Joint)	66.80	0.934	49.651	[34.092, 86.154]	95.67
Portal exit	LeWM	Original → Mixed (Frozen Encoder)	94.34	0.264	11.124	[9.727, 13.437]	89.33
Portal exit	PLDM	Original	51.50	0.996	1067.079	[840.093, 1434.396]	90.56
Portal exit	PLDM	Mixed	79.10	0.128	36.403	[33.836, 39.033]	97.00
Portal exit	PLDM	Original → Mixed (Joint)	61.52	0.708	23.385	[21.534, 25.531]	98.67
Portal exit	PLDM	Original → Mixed (Frozen Encoder)	98.83	0.191	10.203	[9.431, 11.154]	98.67
Portal exit	DINO-WM	Original	56.45	0.996	42.346	[41.223, 43.577]	100.00
Portal exit	DINO-WM	Mixed	100.00	0.142	8.046	[7.960, 8.133]	100.00

H.1. Data Scale

Figure 3a compares LeWM on pusher displacement using the same Test trajectories at both training-data scales. Table 8 reports all three adaptation metrics for these checkpoints. All data-scale comparisons hold training epochs fixed. For pusher displacement, the original-data component determines the mixture length at both 2,048 and 32,768 pairs, giving 3,483 training batches per epoch. The two-stage scheme uses 17,410 Original-pretraining batches followed by 34,830 Mixed batches, compared with 34,830 batches for Mixed training from scratch.

Table 8. LeWM data scaling on pusher displacement (Test). Entries are means over n training runs; pair counts refer to ContextWorld training data.

Scheme	Pairs	n	Accuracy (%)	History win rate (%)	NRE
Mixed	2,048	3	50.00	51.63	1.000
Mixed	32,768	1	50.78	79.10	0.974
Original → Mixed (Joint)	2,048	1	59.57	83.01	0.813
Original → Mixed (Joint)	32,768	1	91.80	99.02	0.215

Table 9 reports the remaining data-scale comparisons on Test. The sample counts describe pairs, not necessarily independent source episodes. Pusher displacement uses 2,048, 8,192, and 32,768 pairs; friction and damping compare 8,192 with 32,768; mass and portal exit compare 2,048 with 32,768. Gripper carry expands from 2,048 to 10,000 source episodes. Portal expansion also broadens initial distances and action magnitudes, so its comparison changes coverage as well as size.

Table 9. Data-scale comparisons on Test. n counts training repetitions contributing to task accuracy and NRE. Planning repetitions can differ and are not counted as additional training runs.

Task	Model	Scheme	Pairs	n	Accuracy (%)	NRE
Pusher displacement	LeWM	Mixed	2,048	3	50.00	1.000
Pusher displacement	LeWM	Mixed	32,768	1	50.78	0.974
Pusher displacement	LeWM	Original → Mixed (Joint)	2,048	1	59.57	0.813
Pusher displacement	LeWM	Original → Mixed (Joint)	8,192	1	83.40	0.377
Pusher displacement	LeWM	Original → Mixed (Joint)	32,768	1	91.80	0.215
Pusher displacement	PLDM	Mixed	2,048	3	50.20	1.005
Pusher displacement	PLDM	Mixed	32,768	1	56.25	0.835
Pusher displacement	DINO-WM	Mixed	2,048	3	93.82	0.507
Pusher displacement	DINO-WM	Mixed	32,768	1	97.66	0.394
Arm mass	LeWM	Mixed	2,048	3	85.42	0.120
Arm mass	LeWM	Mixed	32,768	1	85.74	0.045
Arm mass	PLDM	Mixed	2,048	3	53.12	1.122
Arm mass	PLDM	Mixed	32,768	1	91.02	0.031
Arm mass	DINO-WM	Mixed	2,048	3	51.17	0.999
Arm mass	DINO-WM	Mixed	32,768	1	78.32	0.673
Contact friction	LeWM	Mixed	8,192	3	50.00	1.000
Contact friction	LeWM	Mixed	32,768	1	50.00	1.000
Contact friction	PLDM	Mixed	8,192	3	50.00	1.002
Contact friction	PLDM	Mixed	32,768	1	50.00	1.001
Contact friction	DINO-WM	Mixed	8,192	3	49.93	1.006
Contact friction	DINO-WM	Mixed	32,768	1	84.18	0.648

Task	Model	Scheme	Pairs	n	Accuracy (%)	NRE
Motion damping	LeWM	Mixed	8,192	3	50.20	1.000
Motion damping	LeWM	Mixed	32,768	1	50.00	1.004
Motion damping	PLDM	Mixed	8,192	3	51.24	2.590
Motion damping	PLDM	Mixed	32,768	1	58.20	2.435
Motion damping	DINO-WM	Mixed	8,192	3	49.61	1.031
Motion damping	DINO-WM	Mixed	32,768	1	56.45	1.042
Gripper carry rule	LeWM	Mixed	2,048	3	50.00	1.000
Gripper carry rule	LeWM	Mixed	10,000	1	50.00	1.000
Gripper carry rule	PLDM	Mixed	2,048	3	50.00	1.003
Gripper carry rule	PLDM	Mixed	10,000	1	49.80	0.997
Gripper carry rule	DINO-WM	Mixed	2,048	3	50.78	0.996
Gripper carry rule	DINO-WM	Mixed	10,000	1	60.16	0.892
Portal exit	LeWM	Mixed	2,048	3	51.37	0.934
Portal exit	LeWM	Mixed	32,768	1	50.78	0.922
Portal exit	PLDM	Mixed	2,048	3	69.47	0.271
Portal exit	PLDM	Mixed	32,768	1	72.27	0.128
Portal exit	DINO-WM	Mixed	2,048	3	99.35	0.549
Portal exit	DINO-WM	Mixed	32,768	1	100.00	0.142

H.2. Model Capacity

Both model sizes use the same 32,768-pair datasets, training seed, Mixed initialization, and 10-epoch recipe. The larger model changes the visual backbone from ViT-Tiny to ViT-Base, the representation width from 192 to 768, and predictor depth from six to nine layers. This is a joint expansion of the encoder and predictor, not a predictor-only intervention. Accuracy and NRE use Test; CEM measures planning success in the original environment.

		Accuracy (%)		NRE		CEM (%)	
Task	Model	18M	183M	18M	183M	18M	183M
Contact friction	LeWM	50.00	50.00	1.000	1.000	76.67	74.00
Contact friction	PLDM	50.00	49.80	1.001	1.061	65.67	47.33
Motion damping	LeWM	50.00	50.20	1.004	1.003	78.00	71.33
Motion damping	PLDM	58.20	50.98	2.435	1.768	61.00	50.33

The four endpoint comparisons do not show a consistent improvement from increasing capacity under this recipe. These endpoint comparisons do not exclude gains from retuning the larger models.